

RAG Application Evaluation Report

1. Dataset Overview

The dataset used for this Retrieval-Augmented Generation (RAG) application consists of the **first two volumes of *The Decline and Fall of the Roman Empire* by Edward Gibbon**, comprising a total of **1,446 pages**. The source material was obtained from **Anna's Archive** in PDF format.

This work was chosen due to its long-form historical narrative, dense paragraph structure, and rich hierarchical organization (volumes, chapters, parts), making it well-suited for evaluating retrieval quality and chunking strategies in a RAG system.

2. Data Ingestion

The ingestion process focused on converting the original PDFs into a clean, structured textual format suitable for downstream processing:

- The PDF files were downloaded from Anna's Archive
- Text was extracted from the PDFs
- The extracted text was converted into **Markdown (.md)** format

Markdown was selected to preserve document structure such as headings and sections, which later enabled reliable metadata extraction.

3. Data Normalization

Raw extracted text contained editorial artifacts, inconsistent formatting, and broken paragraph structures that could negatively impact chunking and retrieval. A multi-stage normalization process was applied to improve semantic clarity and structural consistency.

3.1 Heading Structuring

Markdown headings were added to clearly represent the document hierarchy (volume, chapter, part). This ensured that structural boundaries were explicitly encoded and could later be used for metadata extraction and contextual grounding.

3.2 Removal of Formatting Noise

Unnecessary typographical artifacts such as stray asterisks were removed. These artifacts were introduced during PDF extraction and did not contribute to semantic meaning, but could interfere with embedding quality.

3.3 Footnote and Editorial Cleanup

Editorial insertions and footnotes were removed to prevent non-authorial content from polluting the semantic space. These elements often disrupted paragraph flow and introduced context irrelevant to user queries.

3.4 Editorial Number Removal

Inline editorial numbering (e.g., reference markers or annotations) was eliminated. These numbers fragmented sentences and reduced embedding coherence without adding informational value.

3.5 Line-Wrap Correction

Line breaks introduced by PDF formatting were corrected. Sentences broken across lines were rejoined to restore natural linguistic flow, which is critical for producing meaningful embeddings.

3.6 Paragraph Reconstruction

Paragraphs were reconstructed to ensure each paragraph represented a coherent semantic unit. This step was particularly important because downstream chunking relied heavily on paragraph boundaries.

4. Metadata Extraction

After normalization, metadata was extracted from the Markdown structure and associated with each text segment. The following metadata fields were captured:

- Volume
- Chapter
- Part
- Paragraph

Each text segment was stored alongside its metadata in a JSON structure. This enabled traceability, improved interpretability of retrieved context, and facilitated structured evaluation.

5. Chunking Strategy

Chunking was performed using **LangChain's Recursive Text Splitter**.

Rationale

The paragraphs within each part of the text were often extremely large, exceeding practical token limits for retrieval and generation. Recursive chunking allowed:

- Progressive splitting of large paragraphs

- Preservation of semantic coherence
- Controlled chunk sizes suitable for embedding and retrieval

This approach provided better balance between contextual completeness and retrieval precision compared to fixed-size splitting.

6. Embedding and Vector Storage

- **Embedding Model:** OpenAI `text-embedding-3-small`
- **Vector Database:** FAISS

All chunks were embedded and stored in a FAISS vector index to support efficient similarity-based retrieval at query time.

7. RAG Agent Architecture

The RAG system was composed of:

- A FAISS-based retriever for context selection
- An LLM-based answer generator

Language Model

- **Model:** `gpt-4.1-nano`

The agent generates answers strictly grounded in the retrieved context to minimize hallucinations and ensure factual alignment with the source text.

8. Evaluation Framework

Tools Used

- **DeepEval (open-source):** Evaluation of retriever and agent quality
- **LangSmith:** Latency and performance monitoring

Evaluation Metrics

- **Contextual Recall:** Measures whether all information required to answer a question is present in the retrieved context
 - **Contextual Precision:** Measures how highly relevant context is ranked compared to irrelevant context
 - **Contextual Relevancy:** Measures semantic alignment between retrieved context and the query
 - **Answer Relevancy:** Measures how well the final generated answer addresses the user question
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9. Experimental Observations

Observation	Chunk Size	Overlap	Similarity-k	Avg Contextual Recall	Avg Contextual Precision	Avg Contextual Relevancy	Latency (P50 / P99)	Key Insight
Obs-1 (Best)	300	25	8	~1.00	~0.80-0.85	~0.70-0.85	1.38s / 1.78s	Best overall balance of latency, recall, and precision
Obs-2	500	50	5	~1.00	~0.60-0.70	~0.55-0.75	1.63s / 2.55s	Larger chunks reduce precision due to mixed context
Obs-3	1000	100	4	~1.00	~0.70	~0.65	1.53s / 2.37s	Maximizes completeness but over-includes context
Obs-4	400	40	6	~1.00	~0.75-0.80	~0.75-0.85	1.64s / 2.25s	Strong balance, but higher latency than Obs-1

10. Conclusion

Among all evaluated configurations, **Observation 1 (chunk size = 300, overlap = 25, similarity_k = 8)** demonstrated the most favorable overall performance. It achieved perfect contextual recall while maintaining strong contextual precision and relevancy, and it consistently delivered the **lowest latency** across all experiments.

Although larger chunk sizes occasionally improved contextual completeness, they introduced additional noise and increased tail latency. Observation 1 provided the best trade-off between retrieval quality, response accuracy, and system responsiveness.

Based on these results, **Observation 1 is the recommended configuration** for production deployment of this RAG application, particularly for historical question-answering tasks requiring both accuracy and low-latency responses.